

Phase 1: Programming Fundamentals (১-২ সপ্তাহ)

প্রথমে JavaScript নয়, প্রোগ্রামিংয়ের মৌলিক ধারণা শেখান।

- Programming কী?
- Algorithm ও Flowchart
- Variables (`let`, `const`)
- Data Types
- Operators
- Input / Output
- Comments
- Type Conversion

Practice

- Age Calculator
- BMI Calculator
- Simple Calculator

Phase 2: Control Flow (১ সপ্তাহ)

- `if`
- `if...else`
- `else if`
- `switch`
- Ternary Operator

Practice

- Grade Calculator
- Login Validation
- Even/Odd
- Leap Year

Phase 3: Loops (১ সপ্তাহ)

- `for`
- `while`
- `do...while`
- `break`
- `continue`
- Nested Loop

Practice

- Multiplication Table
- Pattern Printing

- Sum of Numbers
- Guess the Number

Phase 4: Functions (১ সপ্তাহ)

- Function Declaration
- Function Expression
- Arrow Function
- Parameters
- Return
- Scope
- Hoisting (Introduction)

Practice

- Calculator Functions
- Temperature Converter
- Prime Number Checker

Phase 5: Arrays (১ সপ্তাহ)

- Create Array
- Access
- Update
- Loop through Array
- Common Methods

- push
- pop
- shift
- unshift
- splice
- slice
- includes
- indexOf

তারপর

- map
- filter
- reduce
- find
- some
- every

Practice

- Todo List (Console)

- Student Result

Phase 6: Objects (১ সপ্তাহ)

- Object
- Nested Object
- Object Methods
- Destructuring
- Spread Operator

Practice

- Student Database
- Product Object

Phase 7: String Methods

- trim
- split
- join
- replace
- includes
- startsWith
- endsWith
- template literals

Phase 8: DOM (সবচেয়ে গুরুত্বপূর্ণ)

এখান থেকেই শিক্ষার্থীরা JavaScript-এর আসল শক্তি বুঝবে।

- DOM কী
- Select Elements
- Change Text
- Change Style
- Attributes
- Create Element
- Remove Element

Practice

- Dark Mode
- Counter
- Live Clock
- Digital Timer

Phase 9: Events

- click

- submit
- keyup
- input
- change
- mouseover

Practice

- Form Validation
- Character Counter
- Image Slider

Phase 10: Advanced JavaScript

- Callback
- setTimeout
- setInterval
- Closures
- Lexical Scope
- this
- bind
- call
- apply

Phase 11: ES6+

- let / const
- Arrow Function
- Destructuring
- Spread
- Rest
- Template Literals
- Modules
- Optional Chaining
- Nullish Coalescing

Phase 12: Asynchronous JavaScript

- Synchronous vs Asynchronous
- Callback
- Promise
- async/await
- fetch API
- JSON

Practice

- Weather App
- Quote Generator
- Random User API

Phase 13: Error Handling

- try
- catch
- throw
- finally

Phase 14: Mini Projects

প্রতিটি টপিক শেষ হলে একটি ছোট প্রজেক্ট দিন।

- Calculator
- Todo App
- Stopwatch
- Quiz App
- Expense Tracker
- Password Generator
- Notes App

Phase 15: Big Projects

- Weather App
- Movie Search
- Shopping Cart
- Chat UI
- Blog Website
- E-commerce Frontend